

Advisory on Regulation of Stem Cell Therapy



Source: PIB

Why in News?

The **Union Ministry of Health and Family Welfare** issued a comprehensive advisory to all States and Union Territories to **strictly regulate stem cell therapy**, mandating that it can be offered as a standard clinical care only for specific disease conditions approved by the Ministry.

- The advisory was issued in compliance with the **Supreme Court's January 2026 judgment in the *Yash Charitable Trust & Ors. v. Union of India* case**, which underscored the need to penalize the unauthorized and unscientific use of stem cell therapies.

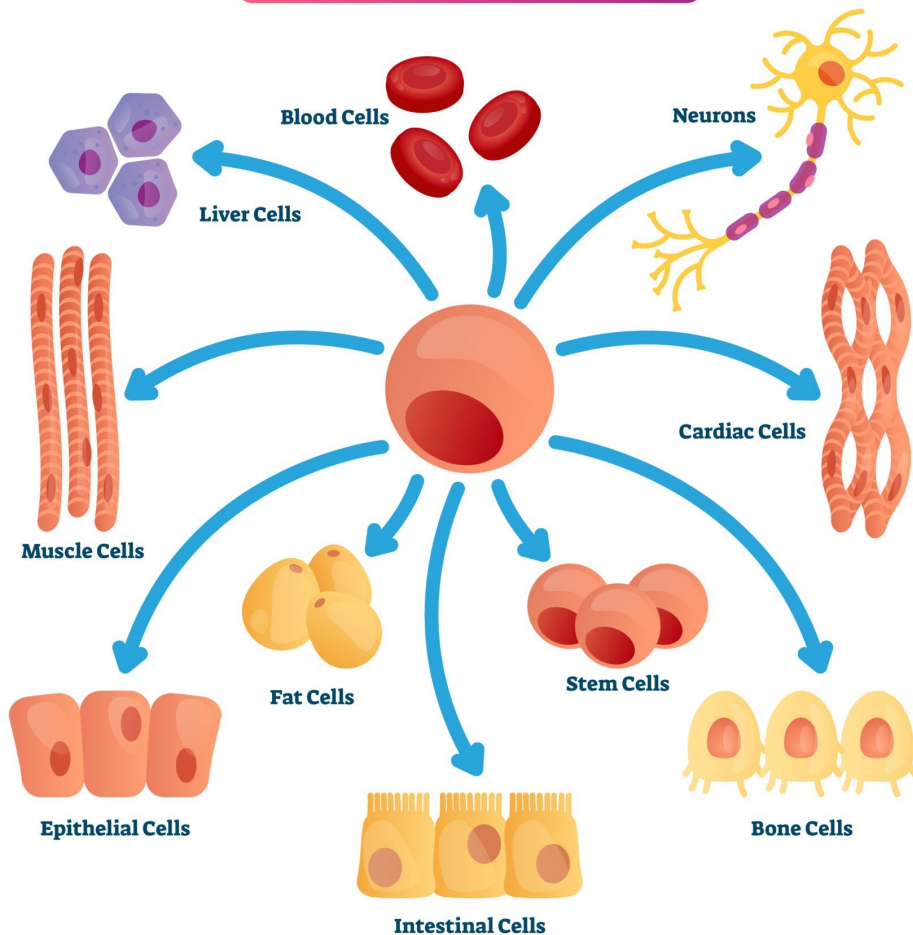
What are the Key Highlights of the Advisory on Regulation of Stem Cell Therapy?

- **Standard Care vs. Experimental Therapy:** Stem cell therapy can be offered as standard care in routine clinical practice **only for disease conditions and indications included in the Ministry-approved list** (e.g., specific blood cancers and haematologic disorders).
- **Restrictions on Autism Spectrum Disorder (ASD):** The therapeutic use of any type of stem cell for **ASD is explicitly restricted to approved clinical trials**.
 - Unproven stem cell interventions for ASD must not be marketed as standard commercial treatments.
- **Compliance with Guidelines:** Any clinical trial involving stem cells must strictly adhere to the **National Guidelines for Stem Cell Research, 2017**, jointly issued by the **Indian Council of Medical Research (ICMR)** and the **Department of Biotechnology (DBT)**.
- **Penalties for Non-Compliance:**
 - The **National Medical Commission (NMC)** reiterated that unauthorized administration, promotion, or advertisement of stem cell therapy beyond approved indications constitutes **professional misconduct** under the **Indian Medical Council (IMC) Regulations, 2002**.
 - Violators will face disciplinary action from State Medical Councils.
 - Establishments may face penalties and cancellation of registration under **Sections 32 and 40 of the Clinical Establishments (Registration and Regulation) Act, 2010**.

What is Stem Cell Therapy?

- **About:** Stem cell therapy, often referred to as **regenerative medicine** , is a clinical treatment that utilizes **stem cells or their derivatives to promote the repair, regeneration, or replacement of diseased, dysfunctional, or injured tissues** .
 - Instead of just treating symptoms with conventional drugs, it harnesses the body's natural cellular healing capabilities.
- **Stem Cells:** Stem cells are the body's foundational "raw materials." Unlike specialized cells (such as skin, muscle, or nerve cells) that have a single fixed purpose, stem cells are **undifferentiated master cells from which all other functional cells are generated**.
- **Core Characteristics:** Stem cells are defined by two unique biological properties:
 - **Self-Renewal:** The ability to **continuously divide and create exact copies** of themselves, maintaining a lifelong reserve pool of healthy cells.
 - **Differentiation:** The capacity to mature and transform into various specialized cell types in response to physiological signals and tissue demands.
- **Types of Stem Cells:**
 - **Embryonic Stem Cells: Pluripotent cells** derived from embryos that can divide into more stem cells or become any type of cell in the body.
 - **Adult Stem Cells:** Found in small numbers in most adult tissues (like bone marrow or fat).
 - They are **typically multipotent**, meaning they can only differentiate into a limited number of cell types.
 - **Induced Pluripotent Stem Cells (iPSCs):** Adult cells that have been genetically reprogrammed to behave like embryonic stem cells.
 - **Perinatal Stem Cells:** Multipotent cells harvested from childbirth-associated tissues, including the placenta, umbilical cord blood, and amniotic fluid.
- **Working Mechanism:** The Stem Cell Therapy process involves **harvesting stem cells** from the patient (**autologous**) or a donor (**allogeneic**), processing them, and administering them through infusion, injection, or implantation.
 - Once introduced, the cells **home to injured tissues, differentiate into needed cell types, and release growth factors and cytokines** that reduce inflammation and stimulate the body's natural repair mechanisms.
- **Current Clinical Landscape:**
 - **Established Medicine:** Haematopoietic stem-cell transplantation (bone marrow transplant) is a globally accepted and established treatment for certain blood cancers (like **Leukemia**) and serious blood disorders (like **Thalassemia**).
 - **Experimental Medicine:** Using stem cells for neurological conditions (like **Autism** , **Alzheimer's** , or **Parkinson's**) is still highly experimental.
- **Key Concerns & Risks:** Stem cell therapy carries risks including **tumour formation, unintended cell differentiation** and immune complications.
 - Embryonic stem-cell research raises ethical concerns, while unregulated clinics may expose patients to serious complications such as infections, embolisms and neurological damage.

STEM CELLS



Regulatory Framework for Stem Cells in India

- **National Guidelines for Stem Cell Research (NGSCR) 2017:** Provides the ethical and scientific framework for stem cell research, categorizing it into permitted, restricted, and prohibited areas.
- **Drugs and Cosmetics Act, 1940 & New Drugs and Clinical Trials Rules, 2019:** Stem cell-derived products (SCDPs) are classified as "New Drugs" and require mandatory approval from the **Central Drugs Standard Control Organisation (CDSCO)** before commercialization.
- **Clinical Establishments Act, 2010:** Regulates hospitals and clinics, ensuring they adhere to minimum standards of facilities and services.
 - The Health Ministry's advisory relies on this Act to mandate registration cancellations for rogue clinics.

Frequently Asked Questions

1. What is stem cell therapy?

Stem cell therapy uses **stem cells or their derivatives** to repair, regenerate or replace damaged or diseased tissues. It is also associated with **regenerative medicine**.

2. Which stem cell therapy is an established medical treatment?

Haematopoietic stem-cell transplantation (bone marrow transplantation) is an established treatment for certain **blood cancers and haematological disorders**, including some cases of leukaemia and thalassemia.

3. Is stem cell therapy for Autism Spectrum Disorder approved as standard care?

No. Stem cell therapy for **ASD is restricted to approved clinical trials** and cannot be marketed as a standard commercial treatment without the requisite approval.

4. What are the National Guidelines for Stem Cell Research, 2017?

Jointly issued by **ICMR and DBT**, the guidelines provide the **ethical and scientific framework** for stem cell research, including categories of permitted, restricted and prohibited research.

5. How are stem cell-derived products regulated in India?

Under the **Drugs and Cosmetics Act, 1940** and **New Drugs and Clinical Trials Rules, 2019**, stem cell-derived products are treated as **new drugs** and require regulatory approval before commercialization.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/Zc-4OqImXCk](https://www.youtube.com/embed/Zc-4OqImXCk)]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims:

Q1. In the context of hereditary diseases, consider the following statements: (2021)

1. Passing on mitochondrial diseases from parent to child can be prevented by mitochondria replacement therapy either before or after in vitro fertilization of the egg.
2. A child inherits mitochondrial diseases entirely from mother and not from father.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Ans: (c)

Q2. With reference to 'stem cells', frequently in the news, which of the following statements is/are correct? (2012)

1. Stem cells can be derived from mammals only
2. Stem cells can be used for screening new drugs
3. Stem cells can be used for medical therapies

Select the correct answer using the codes given below:

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 3 only
- (d) 1, 2 and 3

Ans: (b)

Pradhan Mantri Khanij Kshetra Kalyan Yojana (PMKKKY)



Source: PIB

Why in News?

The **Pradhan Mantri Khanij Kshetra Kalyan Yojana (PMKKKY)** completed significant progress in improving the welfare of mining-affected communities by utilising funds collected through **District Mineral Foundations (DMFs)** .

What are the Key Facts About Pradhan Mantri Khanij Kshetra Kalyan Yojana (PMKKKY)?

- **About:** It was launched on **17th September 2015** to ensure that mining-affected communities receive benefits from mineral extraction through welfare and development projects.
 - The scheme is implemented through **District Mineral Foundations (DMFs)** , which utilise funds collected from mining lease holders.
- **Objectives:** PMKKKY aims to improve the **quality of life in mining-affected areas by developing social infrastructure** , reducing the adverse impacts of mining on the environment, health and livelihoods, and promoting sustainable income opportunities for affected communities.
- **District Mineral Foundation (DMF):** DMFs are non-profit trusts established under the **2015 amendment to the Mines and Minerals (Development and Regulation) Act, 1957** to work for the welfare of people and areas affected by mining operations.
 - They function under State Governments and have been established in **656 districts across 23 States** .
- **Legal Framework:** The **Mines and Minerals (Development and Regulation) Act, 1957** requires States to incorporate PMKKKY provisions into their **DMF rules** . Revised PMKKKY guidelines issued in **January 2024** provide an updated framework for planning and implementing projects funded through DMFs.
- **Protection of Vulnerable Communities:** Since mining activities often affect remote and tribal regions, DMF rules must consider constitutional **safeguards for Scheduled Areas and Scheduled Tribes** , along with provisions of the **Panchayats (Extension to Scheduled Areas) Act, 1996 (PESA)** and **Forest Rights Act, 2006** .
- **Funding Mechanism of DMFs:** DMFs are funded through contributions from mining lease holders:
 - **10% of royalty** for mining leases or prospecting licence-cum-mining leases granted on or after **12 January 2015** .
 - **30% of royalty** for mining leases granted before **12 January 2015** .
- **Fund Utilisation:** At least **70% of PMKKKY funds** are allocated to high-priority sectors such as drinking water, healthcare, education, environmental protection, sanitation, housing, skill development, livelihood generation, agriculture and animal husbandry.
 - Up to **30%** can be used for other priority sectors like infrastructure, irrigation, energy and watershed development.
- **Progress:** DMF-funded projects have expanded significantly, with **over 4.70 lakh projects worth more than ₹1 lakh crore sanctioned** . Of these, more than **2.92 lakh projects have been completed** , while **78,809 projects are under implementation** as of July 2026.

Conclusion

The **Pradhan Mantri Khanij Kshetra Kalyan Yojana (PMKKKY)** represents an important step towards ensuring that communities bearing the social and environmental costs of mining also share its economic benefits. By channelising mineral revenues into healthcare, education, infrastructure

and livelihoods, PMKKKY promotes the idea of **inclusive and sustainable mining-led development** .

Drishti Mains Practice Questions

“PMKKKY seeks to convert mineral wealth into a tool for inclusive development in mining-affected regions.” Discuss the role of District Mineral Foundations in achieving this objective.

Frequently Asked Questions (FAQs)

1. What is PMKKKY?

PMKKKY is a scheme launched in **2015** to promote the welfare and development of mining-affected areas and communities through DMF funds.

2. When was PMKKKY launched?

PMKKKY was launched on **17 September 2015** .

3. What is a District Mineral Foundation (DMF)?

A DMF is a **non-profit trust** established in mining-affected districts for the welfare of affected people and areas.

4. Under which law were DMFs established?

DMFs were introduced through the **2015 amendment to the Mines and Minerals (Development and Regulation) Act, 1957** .

5. Who governs DMFs?

DMFs function under the **jurisdiction of the respective State Governments** .

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/KCVqcAxXoJU](https://www.youtube.com/embed/KCVqcAxXoJU)]

UPSC Civil Services Examination, Previous Years Question (PYQ)

Q. What is/are the purpose/purposes of ‘District Mineral Foundations’ in India? (2016)

1. Promoting mineral exploration activities in mineral-rich districts
2. Protecting the interests of the persons affected by mining operations
3. Authorizing State Governments to issue licences for mineral exploration

Select the correct answer using the code given below:

- (a)** 1 and 2 only
- (b)** 2 only
- (c)** 1 and 3 only
- (d)** 1, 2 and 3

Ans: (b)

Definition of Census Town



Source: TH

The Ministry of Housing and Urban Affairs (MoHUA) has informed a parliamentary panel that the **four-decade-old definition of a Census Town** fails to capture India's actual scale of urbanisation, highlighting the need to replace outdated classification metrics with modern, data-driven approaches.

- **Definition of Census Town:** As per the **1981 Census** , a Census Town is defined as a settlement with a **minimum population of 5,000** , at least **75% of the male working population engaged in non-agricultural pursuits** , and a **population density of at least 400 persons per sq. km** .!
- **Discrepancy in Urbanisation Data:** While the **Census 2011** recorded India's urbanisation at **31.2%** , the **Economic Advisory Council to the Prime Minister (EAC-PM)** estimates that satellite data could put the actual urbanisation level as high as **63% in 2015** .
- **Key Objections Raised by MoHUA:**
 - **Rigid Uniform Thresholds:** The current national parameters disadvantage **hilly and North-Eastern States** .
 - **Male Bias:** The criteria only considers the **male working population** , ignoring female workforce participation.
 - **Flawed Density Calculation:** Population density is calculated using **administrative boundaries rather than built-up areas** , leading to misclassifications.
 - **Outdated Binary:** The rigid **rural-urban binary** overlooks peri-urban growth corridors and regional economic diversity.
- **Registrar General of India (RGI) Stance:** The **RGI** maintained that altering the framework for **Census 2027** is unfeasible as field deployment and enumeration blocks are finalized in advance, though future digital censuses could explore updated parameters.
- **Parliamentary Panel Recommendation:** The Parliamentary Standing Committee on Housing and Urban Affairs deferred the adoption of the draft report and recommended **wider consultations to evolve updated criteria** for future census exercises.

Read more: [Census 2027](#)

India's 1st Soil Carbon Payments Initiative



Source: PIB

Recently, **India's agricultural carbon market** reached the farm level as more than **2,500 smallholder farmers in Punjab and Haryana** received digital payments for adopting **regenerative agricultural practices** under the '**Aadi**' farmer carbon programme .

- **About the Initiative:** The 'Aadi' programme was **launched in 2019 by Grow Indigo** with technical guidance from **Indian Council of Agricultural Research (ICAR)** to link sustainable farming practices with additional farmer income.
- **Farming Practices:** Farmers adopted regenerative practices such as **Direct Seeded Rice (DSR)** , **reduced tillage** , and **crop residue management** between 2019 and 2022 to improve soil health and reduce greenhouse-gas emissions.
 - Farmers received payments based on their share of carbon credits generated from their fields, with the first payments ranging between **₹3,000 and ₹15,000** .
- **Carbon Credit Generation:** The **greenhouse-gas** reductions and increases in soil carbon were scientifically measured and independently verified before carbon credits were issued under the **Verra VM0042 methodology** .
- **Scale of Programme:** The initiative covers over **two million acres and more than 100,000 farmers across seven states** , making it one of India's major agricultural carbon programmes.
- **Environmental Benefits:** The programme estimates that regenerative practices helped save **45 billion litres of water** , prevented more than **two lakh tonnes of crop residue burning** , and avoided around **1,000 tonnes of PM2.5 emissions** .
- **Significance:** The initiative connects **scientific research, carbon markets, and farmer incentives** , showing how regenerative agriculture can support higher farmer incomes, soil improvement, resource conservation, and climate-resilient farming.

Read More: [Regenerative Agriculture for a Resilient India](#)